

Scalable Data Processing API

Solution for Assignment 1 (Out-of-Core Data Join) and Assignment 2 (Non-Blocking FastAPI with two background-processing approaches).

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Assignment 1 — Out-of-Core Data Join: The goal here was to join two large CSV files (5M users and 10M transactions) that are too big to fit in memory at once. The approach uses SQLite as an on-disk buffer — both files are streamed in fixed-size chunks, loaded into temporary tables, indexed on the join key, and then joined using SQL. The result is written back out row by row. Memory usage stays constant regardless of how large the files get.

Assignment 2 — Non-Blocking FastAPI: A REST API that accepts a join request, responds immediately with a job ID, and runs the actual join work in the background. Two approaches are implemented: the first uses FastAPI's *BackgroundTasks* combined with a thread pool (lightweight, no extra setup, works well for I/O-bound jobs); the second uses Python's *ProcessPoolExecutor* to spawn a separate OS process per job, which avoids the GIL entirely and isolates crashes from the main server.

Both assignments share the same core join engine (`join_engine.py`) to avoid duplicating logic. Test data is generated using `generate_data.py`.

